Note on Quantum Hall Effect

Yu Liu

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Abstract

This is a note on QHE, based on David Tong's lecture notes. But I want to make it more concise and capture a smooth idea flow of the topic. My main goal is to basically understand this topic and give insights to its relation with TFT, Anyons and CFT.

Contents

1 Preparations

First we'll give some preparations for the discussion of the QHE. We'll start with some basic concepts in CMT and QM. I won't give a detailed explanation, yet just list out some important results as a reference for later use. Most contents are explained in [1].

1.1 Band Theory and Conductivity

1.1.1 Basic Concepts

In a crystalline solid, the periodic ionic potential causes the allowed electron energies to split into **energy bands**, separated by **band gaps**. The key results are:

Theorem 1.1 **Bloch's Theorem**

The eigenstates of an electron in a periodic potential $V(\boldsymbol{r}+\boldsymbol{R}) = V(\boldsymbol{r})$ take the form

$$\psi_{\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}} u_{\boldsymbol{k}}(\boldsymbol{r}), \tag{1}$$

where $u_{\boldsymbol{k}}$ shares the lattice periodicity $u_{\boldsymbol{k}}(r+R) = u_{\boldsymbol{k}}(r)$and $\boldsymbol{k}$ lives in the first Brillouin zone:

$$\boldsymbol{k} \in \mathrm{BZ} = \left\{ \sum_{i=1}^{d} \alpha_i \boldsymbol{b}_{\boldsymbol{i}} \mid 0 \leq \alpha_i < 1 \right\} \tag{2}$$

where $\boldsymbol{b}_{\boldsymbol{i}}$ are the reciprocal lattice vectors.

Solving the Schrödinger equation with Bloch boundary conditions yields a discrete family of dispersion relations $E_n(\boldsymbol{k})$, labelled by the band index n.

Conceptually, what we do is that we find a symmetry operator T which is the lattice translation operator, and we notice that it commutes with the Hamiltonian, thus we can find a common eigenstate of both H and T to classify the eigenstates into bands.

If the system has a finite number of length for example its a rectangular box L_1, L_2. And we impose suitable boundary conditions, then the allowed $\boldsymbol{k}$ values are discrete, we can calculate how many $\boldsymbol{k}$ states are there in each band, and we find that the number of states in each band is equal to the number of unit cells in the system:

$$N_k = N_c = \frac{L_1 L_2}{a_1 a_2} \tag{3}$$

where a_1, a_2 are the lattice constants. This means that each band can accommodate N_c electrons (ignoring spin).

Here is a rough sketch of the band structure:

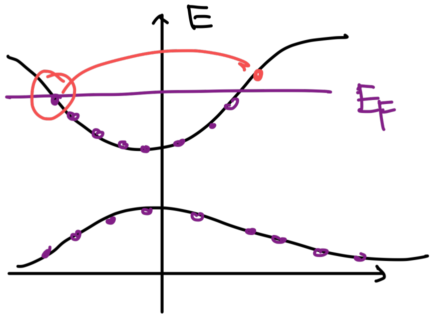

Figure 1: Band structure

Fermi energy and band filling. At $T = 0$, electrons fill all states up to the Fermi energy E_F.

1.1.2 Group Velocity

The group velocity of an electron in a band is given by the gradient of the energy dispersion:

$$v_n(\boldsymbol{k}) = \frac{1}{\hbar} \nabla_{\boldsymbol{k}} E_n(\boldsymbol{k}) \tag{4}$$

This means that the electron's velocity depends on its position in the Brillouin zone and the curvature of the energy band.

This quantity has a physical meaning: if we prepare a wave packet that is a superposition of Bloch states around a certain $\boldsymbol{k}$ point, then the wave packet will move with the group velocity $v_n(\boldsymbol{k})$.

1.1.3 Conductivity

Now let's discuss the conductivity of a material. The key results are:

- **Insulator / Semiconductor:** E_F lies inside a band gap; the valence band is completely full.
- **Metal:** E_F lies inside a band; there exists a Fermi surface.
- **Semimetal:** valence and conduction bands overlap slightly.

1.1.4 Example: Tight Hopping Model

Consider a 2D square lattice with nearest neighbor hopping t and lattice constant a. The tight-binding Hamiltonian is:

$$H = -t \sum_{<i,j>} \left(c_i^\dagger c_j + h.c. \right) \tag{5}$$

There is only one band in this model, and the energy dispersion is:

$$E(\boldsymbol{k}) = -2t\left(\cos(k_x a) + \cos\left(k_y a\right)\right) \tag{6}$$

1.2 Landau Levels

1.2.1 Landau Gauge

Consider a electron moving in a 2D plane under a perpendicular magnetic field B. The Hamiltonian in Landau Gauge is:

$$H = \frac{1}{2m} \left(p_x^2 + \left(p_y + eBx\right)^2 \right) \tag{7}$$

Then the energy spectrum is quantized into **Landau levels**:

- **Wave Functions**

$$\psi_{n,k}(x, y) \sim e^{iky} H_n(x + kl_B^2) e^{-\left(x+kl_B^2\right)^2/2l_B^2} \tag{8}$$

with $H_n \exp\left(-\frac{x^2}{2}\right)$ the n-th Harmonic oscillator wave function and $l_B = \sqrt{\hbar/eB}$ the magnetic length. And $n \in \mathbb{N}$ is the Landau level index, $k \in \mathbb{Z}$ is the momentum along y direction.

- **Energy Spectrum** The energy of the n-th Landau level is:

$$E_n = \left(n + \frac{1}{2} \right) \hbar\omega_B \quad \omega_B = e\frac{B}{m} \tag{9}$$

And each level have a degeneracy of

$$N_B = eBA/2\pi\hbar \quad n = \frac{B}{\Phi_0} \tag{10}$$

which is the number of flux quanta through the system. These energy levels can be drawn in the following CMT style:

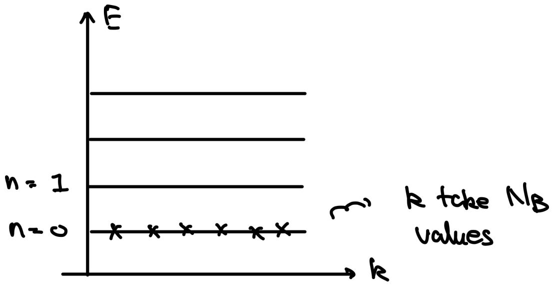

Figure 2: Landau levels

1.2.2 Landau Levels with Electric Field

If we add a uniform electric field E along x direction, then the Hamiltonian becomes:

$$H = \frac{1}{2m}\left(p_x^2 + \left(p_y + eBx\right)^2\right) - eEx \tag{11}$$

Then we solve the Schrödinger equation and find the energy spectrum is still quantized into Landau levels, but with a shift:

- **Wave Functions**

$$\psi(x, y) = \psi_{n,k}(x - mE/eB^2, y) \tag{12}$$

- **Energy Spectrum**

$$E_{n,k} = \hbar\omega_B\left(n + \frac{1}{2}\right) + eE\left(kl_B^2 - \frac{eE}{m\omega_B^2}\right) + \frac{m}{2}\frac{E^2}{B^2} \tag{13}$$

We now see that different k states are not degenerate anymore, and the landau levels are tilted in the energy-momentum space:

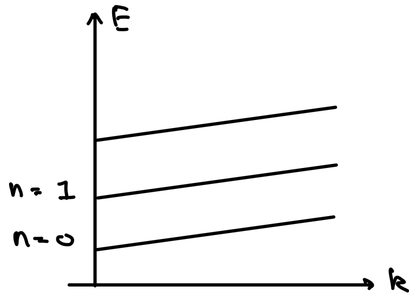

Figure 3: Landau levels with electric field

- **Group Velocity** If the wavefunction is a wave packet that is superposition of the landau level states with different k, then the group velocity of the wave packet is:

$$v_y = \frac{1}{\hbar}\frac{\partial E_{n,k}}{\partial k} = \frac{E}{B} \tag{14}$$

it means that the wave packet will drift along y direction with a velocity.

1.2.3 Symmetry Gauge

Our discussion above is mainly based on Laundau gauge. However, in the discusion of the QHE, especially FQHE (as the original paper does), it's more convenient to use the symmetric gauge.

The symmetry gauge is defined as:

YL: [this part is important.]

1.3 Berry Phase

When a quantum system has a parameter dependent Hamiltonian $H(\lambda_i)$ and the parameters vary adiabatically along a closed loop C in the parameter space, the system's ground state wave function acquires a geometric phase known as the **Berry phase**.

1.3.1 Abelian Berry Phase

Consider the ground state energy is always normalized to 0, no matter which parameter we choose, then after a close loop of parameter variation, the wave function will acquire a phase factor:

$$|\psi(\lambda_i)\rangle \rightarrow e^{i\gamma}\,|\psi(\lambda_i)\rangle \tag{15}$$

where γ is the Berry phase, which can be expressed as an integral of the Berry connection A_i along the loop

- **Berry Connection and Curvature** The Berry connection is defined as:

$$A_i(\lambda) = -i\langle n|\frac{\partial}{\partial\lambda^i}|n\rangle \tag{16}$$

where $|n(\lambda_i)\rangle$ is a series of designated eigenstates of the Hamiltonian with fixed phase choices. And the Berry curvature is defined as:

$$F_{ij}(\lambda) = \partial_i A_j - \partial_j A_i \tag{17}$$

notice that i, j are indices of the parameters, not the spatial coordinates.

- **Gauge Transformation** if we change the choise of phase of the designated eigenstates by a gauge transformation $|n(\lambda_i)\rangle \rightarrow e^{i\alpha(\lambda)}\,|n(\lambda_i)\rangle$, then the Berry connection transforms as:

$$A_i(\lambda) \rightarrow A_i(\lambda) + \partial_i\alpha(\lambda) \tag{18}$$

which is similar to the gauge transformation of the electromagnetic vector potential. However, the Berry curvature is gauge invariant.

- **Berry Phase** can be expressed as the integral of the Berry connection along the loop:

$$i\gamma = -i\oint_C A_i(\lambda)d\lambda^i \tag{19}$$

or equivalently as the integral of the Berry curvature over the area enclosed by the loop:

$$i\gamma = -i\oint_S F_{ij}(\lambda)dS^{ij} \tag{20}$$

- **Chern Number** In a parameter space if we want the berry phase to be well-defined, then the integral of the Berry curvature over a closed surface must be quantized in units of 2π:

$$C = \frac{1}{2\pi}\oint_S F_{ij}(\lambda)dS^{ij} \in \mathbb{Z} \tag{21}$$

this integer C is called the Chern number, which is a topological invariant of the system.

> **Remark**
>
> The quantization of the Chern number is a result of the fact that we want a well-defined holonomy of the Berry connection. This derivation in fact can be generalized to any number of parameters, integrating such a curvature over a closed manifold will give a topological invariant, which is the Chern number.

1.3.2 Non-Abelian Berry Phase

When the system has degenerate ground states, the Berry phase becomes non-Abelian. We can see this from the fact that the designated eigenstates should span the degenerate subspace:

$$|n(\lambda_i)\rangle \to \{|n_a(\lambda_i)\rangle\}_a \tag{22}$$

Then we generalize the definition of Berry connection to a matrix-valued one.

- **Non-Abelian Berry Connection and Curvature** The non-Abelian Berry connection is defined as:

$$A_i(\lambda)_{ab} = -i\langle n_a|\frac{\partial}{\partial \lambda^i}|n_b\rangle \tag{23}$$

and the non-Abelian Berry curvature is defined as:

$$F_{ij}(\lambda) = \partial_i A_j - \partial_j A_i - i\left[A_i, A_j\right] \tag{24}$$

- **Gauge Transformation** if we change the choice of basis of the degenerate subspace by a gauge transformation $|n_a(\lambda_i)\rangle \to U_{ab}(\lambda)\ |n_b(\lambda_i)\rangle$, where $U(\lambda)$ is a unitary matrix, then the non-Abelian Berry connection transforms as:

$$A_i(\lambda) \to U(\lambda)A_i(\lambda)U^\dagger(\lambda) + i(\partial_i U(\lambda))U^\dagger(\lambda) \tag{25}$$

Then we notice that the berry connection is a $U(N)$ gauge field, that take valve of $u(N)$ Lie algebra. And the non-Abelian Berry curvature transforms as:

$$F_{ij}(\lambda) \to U(\lambda)F_{ij}(\lambda)U^\dagger(\lambda) \tag{26}$$

- **Berry Holonomy** The non-Abelian Berry phase (in fact its a matrix) can be expressed as a path-ordered exponential of the Berry connection along the loop:

$$U = P\exp\left(-i\oint_C A_i(\lambda)d\lambda^i\right) \tag{27}$$

1.3.3 Spectrum Flow

1.4 Classical Hall Effect Set Up

Before discussing the non-conventional I/FQHE, let's first talk about the classical Hall effect set up and define useful observable. This is because the QHE effect is in fact kind of based on the classical Hall effect, many description relies on a good understanding of classical Hall effect.

YL: [This part is important. please make sure to understand]

1.4.1 Set Up and Drude Model

1.4.2 Resistivity as Observable

2 Integer Quantum Hall Effect

Now we're prepared to discuss the integer quantum Hall effect (IQHE). The IQHE is a quantum phenomenon observed in two-dimensional electron systems having a quantized Hall conductance. Observation gives us a Hall conductance (resistance) that is quantized:

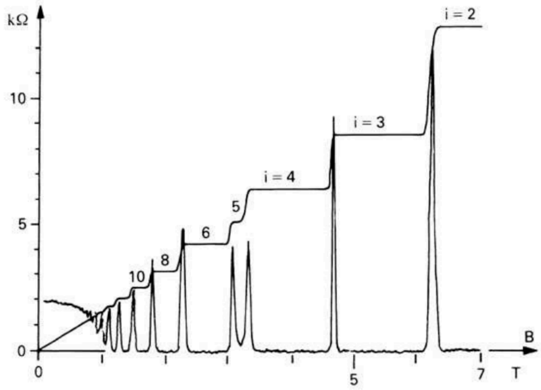

Figure 4: experimental observation of IQHE.

with two important features:

1. **Integer Hall Conductivity**: The Hall conductance σ_{xy} is quantized in integer multiples of:

$$\rho_{xy} = \frac{2\pi\hbar}{e^2}\frac{1}{n}, \quad n = 0, 1, 2, ... \tag{28}$$

1. **Plateau Regions**: The quantized Hall conductance occurs in plateau regions as a function of the magnetic field or electron density, where the conductance remains constant over a range of parameters.
2. **Vanishing Longitudinal Conductivity**: The longitudinal conductance σ_{xx} vanishes in the plateau regions where the Hall conductance is quantized.

> **Note**
>
> Note that in IQHE, in fact ρ_{xx} and σ_{xx} both vanishes. This is not a conflict of being both a perfect conductor and a perfect insulator, it just means that no current is flowing in longitudinal direction, and the system is dissipationless.

2.1 IQHE of Free Electrons

The standard IQHE explanation occurs in a two-dimensional electron gas (2DEG) confined to a plane and subjected to a perpendicular magnetic field. This section we will discuss the explanation of IQHE in this framework. I will not go too much into the details of this, which can be found in many textbooks. Instead, I will just give a brief overview of the key points.

2.1.1 Laughlin's Illustration

Here I'll provide an illustration (I'd prefer to call it an "illustration" for it doesn't really explain a concrete eperiental model, rather illustrate how can this phenonmenon be true) of the IQHE, which is originally given by Laughlin [2].

This illustration is based on a very special geometry:

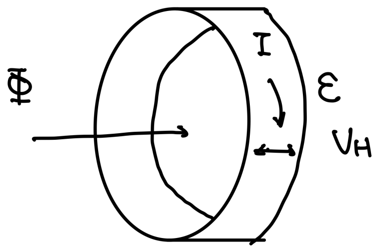

Figure 5: Laughlin's illustration of IQHE geometry.

1. **Integer Conductivity**: The key idea of the IQHE having an integer quantized Hall conductance is that each Landau level go through a spectral flow and thus contribute to one electron going from a edge to another as the flux that generater the electric field goes up by one flux quantum.
2. **Plateau Regions**: The plateau regions can be explained by the presence of disorder in the system, which leads to localized states that do not contribute to the conductance, and they are not effected by the flux. The extended states that contribute to the conductance, and are effected by the flux and may have a spectral flow. As the magnetic field goes down, the localized states are filled first which lead to unchange of the conductance.

For a more detailed discussion, one can see Laughlin's original paper [2], or modern textbooks like [3].

2.1.2 Edge States

Halperin [4], followed Laughlin's illustration, yet he also point out that a negelected but important aspect of IQHE, which is the edge states. Here I follow [1] notes to give an illustration of how the edge states gives out the x direction current with a Hall voltage in y direction.

The picture is that at the edge of the system, there will be a potential that confines the electrons in the bulk. Then the landau levels will bend up due to the energy goes up at the edge.

YL: [To be continued... perhaps put it in the bulk edge section]

2.1.3 A Topological Illustration

In section 12.7 of [3], Fradkin gives a topological illustration of the IQHE. It focus on a geometry of a 2D electron gas on a torus. The key result is that the Hall conductance can be expressed as a topological number of the system. It looks like the TKNN invariant, which we are going to discuss in the next section.

2.2 IQHE in Lattice Models

Apart from the electron gas picture. Sometimes we can't negelect the lattice structure of the system with IQHE, and thus we need to consider to explain the IQHE in a lattice model.

We may see that in fact if the lattice model has a nontrivial structure, we don't even need an external magnetic field to have a quantized Hall conductance. This is the so called quantum anomalous Hall effect (QAHE), which first discovered in [5] experiment.

2.2.1 Kubo Formula

2.2.2 TKNN Formula

2.2.3 Example 1: Hopping Models with Magnetic Field

2.2.4 Example 2: Chern Insulator Models

This is a series of model that can have nontrivial TKNN invariant and thus exhibit quantized Hall conductance without external magnetic field.

The most famous one is the Haldane model, which is a tight-binding model on a honeycomb lattice with complex next-nearest neighbor hopping. Here we might just discussed another simpler model, the Qi-Wu-Zhang (QWZ) model [6].

3 Fractional Quantum Hall Effect

The fractional quantum Hall effect (FQHE) is a quantum phenomenon observed in two-dimensional electron systems subjected to low temperatures and strong magnetic fields, where the Hall conductance exhibits quantized plateaus at fractional values of:

$$\rho_{xy} = \frac{2\pi\hbar}{e^2}\frac{1}{\nu}, \quad \nu = \frac{1}{3}, \frac{2}{5}, \frac{3}{7}, ... \tag{29}$$

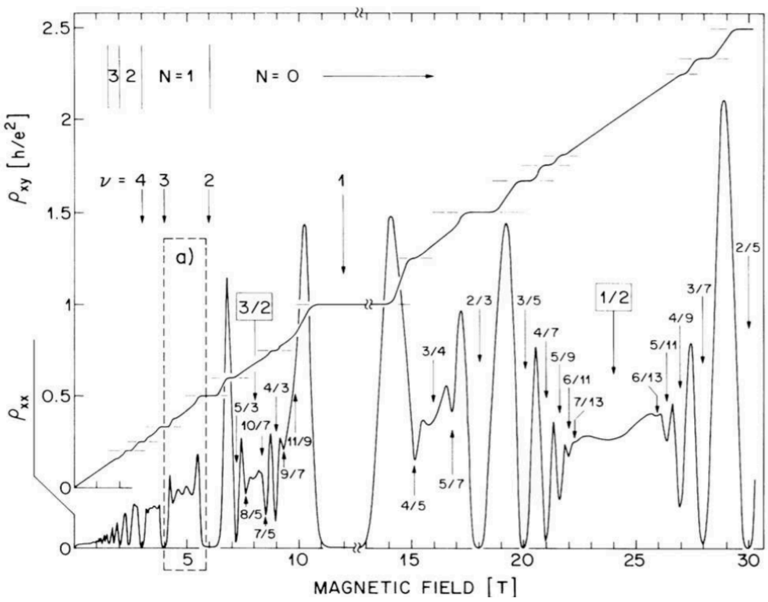

Figure 6: experimental observation of FQHE

[1] The explanation of FQHE is more complicated than IQHE.

3.1 Laughlin States

For FQHE, with the conductivity as:

$$\rho_{xy} = \frac{2\pi\hbar}{e^2}\frac{1}{\nu}, \nu = \frac{1}{m}, m = 3, 5, 7, ... \tag{30}$$

Laughlin give an explanation of this FQHE effect in [7]. What he did is to directly write down a ground state wave function for the state of interacting electrons in magnetic field with the number of:

$$N_e = N_L\nu, \quad N_L = \frac{BA}{2\pi\hbar}, \quad \nu = \frac{1}{m} \tag{31}$$

which has the number of elecrons as a fraction of the number of Landau levels. We often call ν the filling fraction, which is the ratio of the number of electrons to the number of available states in the lowest Landau level.

Now, lets take a look at what this ground state wave function is and why a systems at this state can give rise to a fractional quantum Hall effect.

3.1.1 Laughlin's Wave Function

The Laughlin wave function is given by:

[1] David Tong gives some illustration of how FQHE happens in the beginning, but I think what he wrote is wrong and only make things more confusing.

Theorem 3.1 **Laughlin Wave Function**

For a system of N_e electrons in a magnetic field, the Laughlin wave function at filling fraction $\nu = \frac{1}{m}$ is given by:

$$\psi\left(z_1, z_2, ..., z_{N_e}\right) = \prod_{i<j} \left(z_i - z_j\right)^m \exp\left(-\frac{1}{4l_B^2} \sum_{i=1}^{N_e} |z_i|^2\right) \tag{32}$$

where $z_i = x_i + iy_i$ is the complex coordinate of the i-th electron, $l_B = \sqrt{\frac{\hbar}{eB}}$ is the magnetic length, and m is an odd integer.

This wave function in fact gives a description of a new type of phase of matter called the topological order, which is a type of order that cannot be described by the traditional Landau symmetry breaking theory.

3.2 Excitations: Quasiparticles/Quasiholes

Ground state wave function is not enough to explain the FQHE, we also need to understand the excitations of the system, which are called quasiparticles.

4 Non-Abelian QHE

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